Paired preparations and conditional output fidelity

Exact compatibility of two first-qubit marginals for one outcome-selected operation

Model and criterion

Consider a single nonzero completely positive, trace-nonincreasing operation on two qubits. Its success effect is H, its mean success on the maximally mixed input is $p_\star=\mathrm{Tr}(H)/4$, and $g_i=p_i/p_\star$ for success on inputs $|00\rangle$ and $|01\rangle$ ($i=0,1$). Assume both $p_i>0$. The trace-one Choi state C has reference marginal $H^{\mathsf{T}}/\mathrm{Tr}(H)$. Its population w in the operator subspace $\{L\otimes I_2\}$ is bounded below by w_0. Denote the normalized first-output-qubit states by ρ_0,ρ_1. Write $F=\|\sqrt{\rho_0}\sqrt{\rho_1}\|_1$ for root fidelity.

Fix admissible g_0,g_1 ($g_0+g_1\le 4$) and impose only these relative successes, the two first marginals and $w\ge w_0$. Such a branch exists if and only if

$$g_0+g_1-2\sqrt{g_0g_1}\,F(\rho_0,\rho_1)\le 8(1-w_0).$$

Necessity

Scale the Kraus matrices by $1/\sqrt{p_\star}$ and write their second-qubit blocks as A_j,B_j,C_j,D_j in row order. Projection onto $\{L\otimes I_2\}$ gives $4(1-w)=\Sigma_j[\tfrac12\|A_j-D_j\|^2+\|B_j\|^2+\|C_j\|^2]$ with Frobenius norms. In a common dilation let ψ_0,ψ_1 be the normalized success vectors; flip the second output bit of ψ_1. The squared norm of $\sqrt{g_0}\,\psi_0-\sqrt{g_1}\,\psi_1$ after this alignment is

$$\Sigma_j\,[\|(A_j-D_j)|0\rangle\|^2+\|(C_j-B_j)|0\rangle\|^2]\le 8(1-w).$$

The real part of their overlap is at most $F(\rho_0,\rho_1)$, by the purification characterization of fidelity. Expanding the squared norm proves necessity. The same overlap lower bound applies to any common environmental measurement. The output bit flip is an algebraic comparison of the vectors and leaves the first marginal intact.

Sufficiency

Choose purifications η_0,η_1 of the prescribed first marginals with real overlap F. For the first two input columns use $\sqrt{g_0}\,\eta_0$ with second output 0 and $\sqrt{g_1}\,\eta_1$ with second output 1. For the other columns repeat one normalized first-qubit/environment vector in a disjoint environment subspace, with second outputs 0 and 1 and common norm $\sqrt{[(4-g_0-g_1)/2]}$. The four columns are orthogonal. The projection deficit is exactly the left side divided by eight. Scale all columns together to make the operation trace-nonincreasing. If $p_\star$ is fixed, use its scale whenever all four raw success probabilities lie in $[0,1]$.

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A test using all attempted preparations

Keep the success frequency and both conditional outcomes for each input

Observable probability inequality

Let r_{iy} = Pr(success and binary first-qubit result y | input i), y=0,1, with the same binary effect in both arms. Record unsuccessful attempts as well. Classical root fidelity of the two normalized output distributions is at least $F(\rho_0,\rho_1)$. Multiplying the criterion on page 1 by p★ gives the necessary test

$$\Sigma_y \left(\sqrt{r_{0y}} - \sqrt{r_{1y}}\right)^2 \le 8\, p\star\, (1-w_0).$$

Each rate uses all attempted preparations in its arm as its denominator. For a rank-one projective readout with only these two arm distributions specified, commuting first-qubit states diagonal in its basis attain the classical fidelity. For a general known effect, the displayed test is necessary. Extra measured state or detector constraints can further reduce compatibility.

Sharp difference for the supplied first marginal

At $g_0=g_1=1$, $w_0=0.99$ gives $F \ge c=0.96$. Put $s=\sqrt{(1-c^2)}=0.28$. For a projector whose arm-0 probability is $x<c^2$, the arm-1 probability is at most $G(x)=(c\sqrt{x}+s\sqrt{(1-x)})^2$, by the angle between the binary square-root probability vectors. If the arm-0 larger eigenvalue is $a\approx 0.529472934504$, then $x \in [1-a,a]$ for a rank-one projector.

The maximum of $G(x)-x$ on $[0,c^2]$ is at $x=(1-s)/2$; the difference decreases beyond that point (differentiate the concave function). Here $(1-s)/2=0.36 < 1-a$. Exchanging outcomes bounds the opposite sign. A maximizing trace-distance measurement can be a rank-one projector for two qubit states, so every common binary effect obeys

$$\left|\Pr_1(\text{click} \mid \text{success}) - \Pr_0(\text{click} \mid \text{success})\right| \le 0.272954.$$

The underlying maximum is 0.27295396145309825…. It is attained in the reduced class by diagonal states with event probabilities $x=1-a$ and $G(x)$, using page 1. Attainment under all supplied joint-projector constraints is open.

Equal raw successes can leave the test uninformative

Take orthogonal first marginals, $g_0=g_1=0.04$ and $p\star=0.5$. The page 1 construction gives other relative successes 1.96, and $w=0.99$. The tested inputs both succeed with probability 0.02; the other basis inputs succeed with probability 0.98. A common projector distinguishes the first marginals perfectly. This is a mathematical counterexample to using equality of the two success rates alone. It does not impose the reference value of a.

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Neighboring output and Choi purity

A single population deficit constrains both second-qubit flip probabilities

Equal relative successes at the model point

Keep $g_0=g_1=1$, $w\ge 0.99$ and the supplied value of a. The following bounds allow a common nonzero branch; trace preservation is sufficient. Write Ω_i for its full conditional output and ρ_i for its first marginal. Let β,α be the conditional probabilities of a second-bit flip on inputs 0,1 respectively. The projection identity on page 1 yields $\beta+\alpha \le 4(1-w) \le 0.04$. A bound from the first marginal and the measured second-bit population gives $\lambda\max(\Omega_0) \le a+\beta$ and $\lambda\max(\Omega_1) \le \lambda\max(\rho_1)+\alpha$.

For clarity, a positive two-by-two output block matrix with diagonal blocks A,D has largest eigenvalue $\le \|A\|+\|D\|$ by block positivity and Cauchy–Schwarz. The first diagonal block is bounded by the first marginal; the other by the flip probability. Exchange the two blocks on input 1. Page 2 gives $\lambda\max(\rho_1) \le G(a) < 0.793184183697$ by testing the top projector of ρ_1; G is increasing up to c^2.

Upper limit for the unmeasured joint output

Apply the binary probability bound G to the environment projector for its largest arm-1 eigenvalue. On arm 0 this projector has probability $\le a+\beta$; the environment and full joint output have the same nonzero spectra on each successful pure input. Here $a+\beta < c^2$. Consequently,

$$\lambda\max(\Omega_1) \le \min\{\, G(a+\beta),\, G(a)+0.04-\beta \,\} < 0.810954608659.$$

Split at $b=0.022229575038$. For $\beta\le b$ use $G(a+b)$; for $\beta\ge b$ use $G(a)+0.04-b$. Both are below the displayed decimal by outward interval arithmetic. The submitted single-arm comparison gave 0.824617080524.

Upper limit for normalized Choi purity

The environment marginal purifying C is the sum of the four basis-input environment states weighted by $g_i/4$; a zero-success input contributes zero. The two weights considered here are 1/4 each and the remaining weights sum to 1/2. For every environment ray, the first two probabilities sum to at most $a+\beta+G(a)+\alpha$. Thus $\lambda\max(C) \le t=[2+a+G(a)+0.04]/4$. Since $t\ge ½$, the spectrum $(t,1-t,0,\ldots)$ majorizes the spectrum of C, yielding

$$\mathrm{Tr}(C^2) \le t^2+(1-t)^2 < 0.732104302723.$$

Both upper bounds use the shared flip budget without optimizing every compatibility constraint. Their sharp values remain open. The two-input distance lower bound 0.173002789690 in the submitted report concerns unitary Choi states in the specified channel model.

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